Problem 1

$$\hat{SE}(\bar{X}) = \frac{S}{\sqrt{n}} = \frac{10}{\sqrt{100}} = 1$$

$$z = \frac{\bar{X} - \mu_0}{\hat{SE}(\bar{X})} = \frac{52-50}{1} = 2$$

$\alpha = 0.10$, reject H_0 since $z = 2 > 1.645$ 90% CL

$\alpha = 0.05$, reject H_0 since $z = 2 > 1.96$ 95% CL

$\alpha = 0.01$, fail to reject H_0 since $z = 2 < 2.576$. 99% CL

A 95% CI for μ is

$$\bar{X} \pm 1.96\,\hat{SE}(\bar{X}) = 52 \pm 1.96 \times 1 = (50.04, 53.96)$$

Problem 2

90% CI: $3.1 \pm 1.645 \times 1.3 = (0.9615, 5.2385)$

95% CI: $3.1 \pm 1.96 \times 1.3 = (0.552, 5.648)$

99% CI: $3.1 \pm 2.576 \times 1.3 = (-0.2488, 6.4488)$

$H_0: \sigma = 0$ vs. $H_1: \sigma \neq 0$

$$z = \frac{\hat{\sigma} - \sigma_0}{\hat{SE}(\hat{\sigma})} = \frac{3.1 - 0}{1.3} \approx 2.3846$$

$\alpha = 0.10$, $z > 1.645$, reject H_0

$\alpha = 0.05$, $z > 1.96$, reject H_0

$\alpha = 0.01$, $z < 2.576$, fail to reject H_0

Problem 3

$$\hat{SE}(\bar{X}) = \frac{S}{\sqrt{n}} = \frac{3}{\sqrt{36}} = 0.5$$

95% CI: $\bar{X} \pm 1.96 \times 0.5 = 11.5 \pm 1.96 \times 0.5 = (10.52, 12.48)$

$\mu_0 = 12 \in (10.52, 12.48)$ fail to reject H_0.

$$z = \frac{\bar{X} - \mu_0}{\hat{SE}(\bar{X})} = \frac{11.5 - 12}{0.5} = -1$$

$\alpha = 0.05$: $|z| = 1 < 1.96$, fail to reject H_0.

$$p = P(Z \leq -1) + P(Z \geq 1) = 2F(-1) = 2(1 - F(1))$$

$$= 2 \times (1 - 0.8413) = 0.3174 > 0.05$$, fail to reject H_0.

Problem 4

$$\hat{SE}(\bar{X}_A-\bar{X}_B)=\sqrt{\frac{S_A^2}{n_A}+\frac{S_B^2}{n_B}}=\sqrt{\frac{8^2}{100}+\frac{7^2}{100}}\approx 1.063$$

$H_0: \mu_A-\mu_B=0$ vs. $H_1: \mu_A-\mu_B\neq 0$

$$z=\frac{(\bar{X}_A-\bar{X}_B)-0}{\hat{SE}(\bar{X}_A-\bar{X}_B)}=\frac{25-22}{1.063}\approx 2.8222$$

$\alpha=0.05$: $|z|>1.96$, reject H_0

95% CI: $(\bar{X}_A-\bar{X}_B)\pm 1.96\sqrt{\frac{S_A^2}{n_A}+\frac{S_B^2}{n_B}}=3\pm 1.96\times 1.063$

$=(0.9165, 5.0835)$

Since CI excludes zero, reject H_0

Problem 5

$$\hat{SE}(\bar{X}_A-\bar{X}_B)=\sqrt{\frac{S_A^2}{n_A}+\frac{S_B^2}{n_B}}=\sqrt{\frac{6^2}{81}+\frac{5^2}{64}}\approx 0.9138$$

$H_0: \mu_A-\mu_B=0$, $H_1=\mu_A-\mu_B\neq 0$

$$z=\frac{(\bar{X}_A-\bar{X}_B)-0}{\hat{SE}(\bar{X}_A-\bar{X}_B)}=\frac{(14-12)-0}{0.9138}\approx 2.1886$$

Problem 6

$$\hat{SE}(\bar{X}_A-\bar{X}_B)=\sqrt{\frac{S_A^2}{n_A}+\frac{S_B^2}{n_B}}=\sqrt{\frac{1.5^2}{49}+\frac{1.4^2}{64}}\approx 0.2767$$

$H_0: \mu_A-\mu_B=0$, $H_1=\mu_A-\mu_B\neq 0$

$$z=\frac{(\bar{X}_A-\bar{X}_B)-0}{\hat{SE}(\bar{X}_A-\bar{X}_B)}=\frac{(3.2-2.8)-0}{0.2767}\approx 1.4458$$

$\alpha=0.05$, $|z|=1.4458<1.96$, fail to reject H_0

$p=P(Z\leq -1.4458)+P(Z\geq 1.4458)=2F(-1.4458)$

$=2(1-F(1.4458))=2\times(1-0.926)=0.148>0.05$

$p>\alpha$, fail to reject H_0.

The observed difference of 0.4 is approximately 1.45 standard errors from zero and is not statistically significant at

at the 5% significance level the 5% level. There is insufficient evidence that the two factories have different population mean defect scores. This does not mean that their population means are equal.